Practice Set 7 Paper 2 – Urology

Contents

1. Q1. Evaluation and management of a 38-year-old man with uncontrolled hypertension (LAQ)
2. Q2. Etiopathogenesis of stone formation and medical management of urolithiasis (LAQ)
3. Q3a. Management of T1G3 bladder tumour (SAQ)
4. Q3b. Organ preservation solution (SAQ)
5. Q3c. Intraoperative management of phaeochromocytoma (SAQ)
6. Q3d. Saturation biopsy in carcinoma prostate (SAQ)
7. Q3e. Steinstrasse (SAQ)
8. Q3f. Renal angiomyolipoma (SAQ)

Practice paper compiled from the question bank.

Q1. How will you evaluate and manage 38 yr male with uncontrolled hypertension

Type: LAQ (25 marks) (renovascular hypertension also answered in Practice Set 2 Paper 2, Q1; phaeochromocytoma in Practice Set 6 Paper 1, Q1)

Assumed focus: secondary (urological, renal and adrenal) causes of hypertension in a young adult.

Introduction

- **Resistant hypertension:** BP (blood pressure) above goal despite 3 drugs of different classes, including a diuretic, at optimal doses; or needing 4 or more drugs.
- Poor control below 40 years is a classic trigger to search for a **secondary cause**; many are curable.

Causes

Group	Causes	Clues
Renovascular	FMD (fibromuscular dysplasia), atherosclerotic RAS (renal artery stenosis), Takayasu arteritis, reninoma	Bruit; creatinine rise >30% on ACE (angiotensin-converting enzyme) inhibitor; flash pulmonary oedema; kidneys differing >1.5 cm
Renal parenchymal	CKD (chronic kidney disease), ADPKD (autosomal dominant polycystic kidney disease), reflux nephropathy, glomerulonephritis, Page kidney	Raised creatinine, proteinuria, haematuria
Obstructive uropathy	Bilateral obstruction (about 77% hypertensive), unilateral (about 20%), chronic retention	LUTS (lower urinary tract symptoms), palpable bladder, hydronephrosis
Primary aldosteronism (PA)	Adenoma (Conn), bilateral hyperplasia	Hypokalaemia (present in <40% only)
Phaeochromocytoma, paraganglioma	Sporadic or hereditary	Paroxysmal headache, sweating, palpitations
Cushing syndrome	Exogenous steroid, adrenal adenoma, pituitary	Central obesity, striae, diabetes

- **Also know:** sleep apnoea; drugs (NSAIDs [non-steroidal anti-inflammatory drugs], steroids, decongestants, liquorice); alcohol; thyroid and parathyroid disease; coarctation.

Evaluation

Step 1: Exclude pseudo-resistance

- Correct cuff technique; home or ambulatory monitoring to exclude white-coat effect; check adherence, doses, interfering drugs, salt and alcohol.

Step 2: History and examination

- Paroxysms, weakness or polyuria (hypokalaemia), stones, UTI (urinary tract infection) or childhood reflux, snoring, family history.
- BP both arms and radio-femoral delay; abdominal bruit; palpable kidneys or bladder; cushingoid habitus; café-au-lait spots (NF1 [neurofibromatosis type 1]); fundus.

Step 3: Baseline tests and target-organ damage

- Creatinine, eGFR (estimated glomerular filtration rate), electrolytes, urinalysis with albumin–creatinine ratio, glucose, lipids, ECG (electrocardiogram), echocardiography.

- **Ultrasound KUB (kidneys, ureters, bladder):** renal size and asymmetry, scars, cysts, masses, hydronephrosis, residual urine.

Step 4: Targeted screening (method and interpretation)

Cause	Screen	Confirm or localise
PA	Morning **ARR (aldosterone-to-renin ratio)** >20–30 with aldosterone ≥15–20 ng/dL; stop spironolactone first; correct potassium	**Intravenous saline 2 L over 4 h** (aldosterone stays >5 ng/dL) or oral salt loading → adrenal CT (computed tomography) → **AVS (adrenal vein sampling)**; skip AVS if age <40 with a clear unilateral adenoma >1 cm
Phaeochromocytoma	**Plasma free or 24-h urinary fractionated metanephrines**	CT or MRI (magnetic resonance imaging); ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); genetic testing
Cushing	**1-mg overnight dexamethasone suppression** (cortisol ≤1.8 µg/dL excludes); late-night salivary cortisol; 24-h urinary free cortisol	ACTH (adrenocorticotropic hormone): low = adrenal, high = pituitary or ectopic
Renovascular	**Duplex Doppler**: peak systolic velocity >180–200 cm/s, renal/aortic ratio >3.5	CT or MR angiography; catheter angiography with pressure gradient if intervention planned
Parenchymal, obstructive	Ultrasound, urine tests	Renogram (DTPA [diethylenetriaminepentaacetic acid] or MAG3 [mercaptoacetyltriglycine]) for split function and drainage; CT urography

```
Uncontrolled BP, age 38 → exclude pseudo-resistance
 → baseline tests + ultrasound KUB
 → ARR, metanephrines, dexamethasone test, renal Doppler
    ├─ positive → confirm → localise → treat cause
    └─ all negative → essential: optimise drugs
```

Management

1. General (all patients)

- Lifestyle: salt restriction, weight loss, exercise, limit alcohol.
- Optimise **ACE inhibitor or ARB (angiotensin receptor blocker) + calcium channel blocker + thiazide-like diuretic**. Fourth drug: **spironolactone** (PATHWAY-2, 2015: best add-on in resistant hypertension).

2. Cause-specific

Cause	Treatment
PA, lateralised	**Laparoscopic adrenalectomy**: biochemical cure about 94%, but only 37–52% need no antihypertensives
PA, bilateral or unfit	Spironolactone 25–50 mg/day (up to 400 mg) or eplerenone
Phaeochromocytoma	**α-blockade 7–14 days**, β-blocker only after α, volume expansion → laparoscopic adrenalectomy; metanephrines yearly for life
Adrenal Cushing	Adrenalectomy with steroid cover; pituitary disease → trans-sphenoidal surgery
FMD	**Angioplasty alone** (cure common in the young)
Atherosclerotic RAS	Medical therapy (**ASTRAL 2009, CORAL 2014**: stenting no benefit); stent for flash pulmonary oedema or falling function

Cause	Treatment
Branch disease, aneurysm	Surgical bypass or ex vivo repair with autotransplantation
Reninoma	Partial nephrectomy (cures most)
Small non-functioning kidney (reflux, segmental hypoplasia)	**Nephrectomy** if renal vein renin lateralises (ratio >1.5)
Page kidney	Drain or decapsulate
Obstruction	**Relieve it** (catheter, nephrostomy, pyeloplasty, prostate surgery); watch post-obstructive diuresis
CKD	ACE inhibitor or ARB for proteinuria; scheduled diuretics

3. Follow-up: home BP, creatinine, potassium; duplex after angioplasty.

Recent advances

- **Catheter renal denervation** (radiofrequency or ultrasound), approved in the USA in 2023 for selected resistant hypertension; SPARTACUS trial questioned routine AVS.

Diagram to draw: stepwise algorithm from pseudo-resistance to cause-specific treatment.

> **Key points to remember:**
> - Hypertension below 40 years: screen for secondary causes.
> - ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
> - Metanephrines for phaeochromocytoma; α before β.
> - FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Q2. Describe etiopathogenesis of stone formation and medical management of urolithiasis

Type: LAQ (25 marks) (also answered in 2020-Summer Paper 4, Q2)

Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to 50% within 5 years. Removal is surgical; recurrence is metabolic.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20%, uric acid and struvite about 7% each, cystine 1–3%.

Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Anatomical**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	< Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors work. **Zone of therapy**
Unstable	> Kf	Spontaneous nucleation, growth, aggregation

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

Steps of stone formation

1. **Supersaturation** (low volume, high calcium, oxalate, uric acid; low citrate).
2. **Nucleation**, then **growth** and **aggregation** (aggregation can occlude a tubule within the 5–7 minute nephron transit).
3. **Retention**: fixed-particle theory. **Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) suggest a secondary cause (enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria).

Inhibitors, promoters, matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm-Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

Pathogenesis by abnormality

Abnormality	Mechanism
Hypercalciuria (commonest)	Absorptive (gut), renal leak (high PTH (parathyroid hormone)), resorptive (primary hyperparathyroidism)
Hyperoxaluria	Primary (hepatic enzyme defect, PH1 AGXT), **enteric** (fat binds calcium, freeing oxalate), dietary
Hyperuricosuria	Urate promotes calcium oxalate nucleation
Hypocitraturia	Distal RTA, diarrhoea, hypokalaemia, animal protein (intracellular acidosis)
Uric acid	**Low urine pH** (<5.5; pKa 5.5) more than excess uric acid
Cystine	Autosomal recessive defect in renal cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate + carbonate apatite; biofilm

Medical management

1. Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm**: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

2. Dissolution: uric acid stones (radiolucent) dissolve with oral potassium citrate (EAU (European Association of Urology): pH 7.0–7.2 for dissolution).

3. Prevention: general (every former)

- **Fluid** for urine **≥2.5 L/day** (Borghi 1996: 12.1% vs 27% recurrence at 5 years).
- **Normal calcium**, low sodium, less animal protein, more fruit and vegetables (Borghi 2002: halved recurrence versus low-calcium diet).

4. Metabolic evaluation (high-risk and recurrent formers): stone analysis, serum chemistry, one or two **24-h urines**; repeat within 6 months of therapy, then yearly.

5. Selective drug therapy

Abnormality	Drug and dose	Rationale
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption; add potassium citrate (hypokalaemia → hypocitraturia)
Hypocitraturia, distal RTA	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular citrate reabsorption
Hyperuricosuria	Allopurinol 300 mg/day	Watch hypersensitivity
Uric acid stones	Potassium citrate to **pH 6.0–6.5**	pH disease
Enteric hyperoxaluria	**Calcium with meals**, low oxalate and fat, magnesium	Binds gut oxalate
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA (ribonucleic acid) interference)	ILLUMINATE-A
Cystinuria	Urine ≥3 L, pH **7.0–7.5**; **tiopronin**	Cystine <250 mg/L; mixed disulfide
Struvite	Complete clearance; AHA (acetohydroxamic acid) 250 mg three times daily	Urease inhibitor; thromboembolism

- **Also know:** recurrent calcium stones with normal work-up: thiazide or citrate (AUA (American Urological Association)); NOSTONE (2023): hydrochlorothiazide did not reduce recurrence; citrus juices are adjuncts.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a papilla.

Key points to remember:

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L; MET for distal 5–10 mm stones.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5, <250 mg/L.

Q3a. Discuss management of T1G3 bladder tumor

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 2, Q1)

Introduction

- T1G3: high-grade (HG) urothelial carcinoma invading the lamina propria, not muscle. It is **high-risk** NMIBC (non-muscle-invasive bladder cancer): recurrence >80%, progression 30–50%.
- **Adverse factors:** concurrent CIS (carcinoma in situ), LVI (lymphovascular invasion), variant histology, size >3 cm, multifocality, prostatic urethral involvement, residual T1 at re-resection.

Step 1: Complete TURBT (transurethral resection of bladder tumour)

- Resect all tumour **with detrusor muscle**; en bloc where possible; blue-light or NBI (narrow-band imaging) for CIS; CT (computed tomography) urography.

Step 2: Re-TURBT at 2–6 weeks (all T1)

- Residual tumour in ≥40%; upstaging to muscle invasion in 25–30%. Improves recurrence- and progression-free survival.

```
HG T1 → re-TURBT
   ├─ ≥T2                → neoadjuvant chemotherapy + cystectomy
   ├─ very high risk     → offer early radical cystectomy
   └─ T0, Ta, CIS, T1    → BCG (bacille Calmette-Guérin)
```

Step 3: BCG – standard

- **Th1 (T-helper 1)** immune response; the **only intravesical agent that reduces progression**.
- Induction weekly ×6, 2–4 weeks after TURBT. **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000)**: 3 weekly doses at 3 and 6 months, then 6-monthly to **3 years**.
- Avoid with gross haematuria, traumatic catheterisation or UTI (urinary tract infection).

Step 4: Early radical cystectomy

- Clinical T1 is understaged in one-third to one-half; progressors do worse than primary muscle-invasive disease.
- Offer for **very high risk** (EAU [European Association of Urology]): T1 HG with CIS plus risk factors (age >70, multiple tumours, ≥3 cm), LVI, variant histology, prostatic urethral CIS, persistent T1 at re-TURBT.

Step 5: BCG-unresponsive disease

- HG T1 at 3 months, or recurrence within 6 months (T1) or 12 months (CIS) of adequate BCG.
- **Radical cystectomy**; if unfit: pembrolizumab (KEYNOTE-057), nadofaragene firadenovec, gemcitabine–docetaxel, trials.

Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly, then yearly; upper tracts yearly.

Diagram to draw: bladder wall layers with Ta, T1 and T2.

Key points to remember:

- Re-TURBT in all T1 at 2–6 weeks.
- BCG with 3-year maintenance reduces progression.
- Very high risk or BCG-unresponsive: radical cystectomy.

Q3b. Organ preservation solution

Type: SAQ (10 marks)

Assumed context: preservation of the kidney for transplantation.

Principles

- **Hypothermia (4 °C)** slows metabolism to roughly a tenth but does not stop it, so injury accumulates with time.
- Ischaemia causes ATP (adenosine triphosphate) depletion, Na^+/K^+ pump failure, **cell swelling**, **acidosis** and calcium influx. At reperfusion, hypoxanthine is converted by xanthine oxidase to **free radicals** (ischaemia–reperfusion injury). Clinically this appears as **DGF (delayed graft function)**.
- A solution therefore needs:
  - **Impermeants** (lactobionate, raffinose, mannitol, glucose) to prevent cell swelling.
  - **Buffers** (phosphate, histidine) against acidosis.
  - **Colloid** (hydroxyethyl starch, polyethylene glycol) against interstitial oedema.
  - **Antioxidants and energy precursors** (glutathione, allopurinol, adenosine).
  - Electrolytes: "intracellular" (high K^+, low Na^+) or "extracellular" (low K^+, high Na^+) type.

Solutions

Solution	Type and key components	Points
UW (University of Wisconsin, Belzer)	Intracellular; lactobionate, raffinose, hydroxyethyl starch, glutathione, adenosine, allopurinol	Standard for static storage; viscous; flush out high K^+ before reperfusion
HTK (histidine–tryptophan–ketoglutarate, Custodiol)	Low K^+; histidine buffer, tryptophan (membrane stabiliser), ketoglutarate (substrate), mannitol	Low viscosity, fast cooling; large volume; outcomes similar to UW
Euro-Collins	Intracellular; glucose impermeant, phosphate	First generation; largely abandoned
Celsior	Extracellular; lactobionate, mannitol, glutathione, histidine	Designed for heart; also abdominal organs
IGL-1 (Institut Georges Lopez)	UW-like, but polyethylene glycol and reversed Na^+/K^+	Extracellular; less viscous

- **Also know:** Marshall hypertonic citrate; cold lactated Ringer in living donation (very short cold ischaemia).

Methods

- **Static cold storage:** in situ aortic flush, IVC (inferior vena cava) venting, ice slush; storage at 4 °C. Aim cold ischaemia <24 hours (thrombosis and DGF rise beyond).
- **Hypothermic machine perfusion:** pulsatile cold perfusion; flow and resistance help assess viability. Reduces DGF; **Moers (2009)**: DGF 20.8% vs 26.5%, 1-year graft survival 94% vs 90%. The chapter notes no difference in long-term survival. Used especially for DCD (donation after circulatory death) kidneys.
- **Normothermic perfusion:** allows assessment and repair; limited experience.

Diagram to draw: ischaemia–reperfusion cascade with the solution component that blocks each step.

> **Key points to remember:**
> - Hypothermia slows, not stops, metabolism.
> - UW: lactobionate, raffinose, starch; HTK: histidine, tryptophan, ketoglutarate.
> - Machine perfusion reduces DGF.

Q3c. Intra op mgt of pheo.

Type: SAQ (10 marks) (preoperative management also answered in 2020-Summer Paper 2, Q3c; perioperative in Practice Set 6 Paper 1, Q1)

Why it is difficult

- Phaeochromocytoma surgery is a **two-phase haemodynamic problem**: catecholamine **surge** while the tumour is handled, then catecholamine **withdrawal** once the vein is tied (hypotension, hypoglycaemia).
- Surges occur at intubation, positioning, pneumoperitoneum and tumour manipulation.

Prerequisites

- **α-blockade 7–14 days** (phenoxybenzamine or doxazosin), β-blocker only after α, volume expansion; ECG (electrocardiogram) and echocardiography; joint plan with the anaesthetist.

Monitoring

- **Arterial line before induction**, central venous line, ECG, urine output, frequent glucose; cardiac output or transoesophageal echo if cardiomyopathy.

Anaesthesia

- Deep, smooth induction (propofol, opioid such as fentanyl or remifentanil); attenuate intubation response.
- **Avoid triggers:** histamine-releasing drugs (morphine, atracurium), ketamine, ephedrine, metoclopramide, droperidol, desflurane.
- Sevoflurane or isoflurane; vecuronium or rocuronium; magnesium sulphate as an adjunct (reduces catecholamine release and arrhythmia).

Haemodynamic control

Event	Action
Hypertensive surge	**Pause manipulation**; nitroprusside, nicardipine or clevidipine, nitroglycerin, phentolamine
Tachyarrhythmia	**Esmolol**, lidocaine; amiodarone if needed
Hypotension after vein ligation	**Aggressive volume**, then α-agonist (phenylephrine, noradrenaline); vasopressin if refractory
Hypoglycaemia (insulin rebound)	Dextrose infusion, glucose checks

Surgical principles

- **Laparoscopic adrenalectomy** (transperitoneal or posterior retroperitoneoscopic) is standard; open for large or invasive tumours.
- **Minimal handling**; "no-touch" dissection away from the tumour.
- **Early adrenal vein ligation** where possible, after warning the anaesthetist.
- Remove intact in a bag (avoid capsule rupture and seeding).
- **Cortical-sparing** (partial) adrenalectomy in MEN2 (multiple endocrine neoplasia type 2), VHL (von Hippel–Lindau) and bilateral tumours.
- Steroid cover if bilateral adrenalectomy.

Postoperative

- HDU (high dependency unit): blood pressure, heart rate, glucose; metanephrines after surgery, then yearly for life (recurrence up to 16%).

Diagram to draw: intraoperative BP curve: surges during handling, fall after vein ligation, with drugs marked.

> **Key points to remember:**
>
> - Arterial line before induction; avoid histamine releasers and ephedrine.
> - Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
> - After vein ligation: fluids, vasopressors, glucose.

Q3d. Saturation biopsy in CA prostate

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 4, Q3a)

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often 20–30), sampling the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone; the 12-core scheme is today's standard.
- Saturation reduces sampling error, mainly for **anterior and apical cancers** missed transrectally.

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation; **focal therapy** mapping (shows multifocality; TRUS [transrectal ultrasound] biopsy misses almost half of disease relevant to hemiablation).

Templates

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine).

Results

- **Initial biopsy**: 12 to 18–21 cores did not raise detection.
- **Repeat setting**: early series gained; a later large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention**; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** detected more significant cancer (38% vs 26%) and 28% avoided biopsy.
- Still used for MRI-negative high suspicion and focal therapy planning.

Diagram to draw: transperineal grid over the prostate with sampling sectors.

Key points to remember:

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Q3e. Steinstresse.

Type: SAQ (10 marks)

Assumed meaning: steinstrasse.

Definition

- **Steinstrasse** (German, "stone street"): a column of stone fragments in the ureter that fails to pass and may obstruct. Typically after **SWL (shock wave lithotripsy)**; occasionally after PCNL (percutaneous nephrolithotomy) or URS (ureteroscopy).
- Incidence is low overall (about 4–7% after SWL, EAU [European Association of Urology]) but rises several-fold with stone burden **>2 cm²**.

Risk factors

- Large burden (>2 cm), renal pelvic stones, hard stones that break into large fragments, high energy, impaired ureteric drainage.

Classification (Coptcoat)

Type	Composition
I	Small fragments only
II	Larger lead fragment (4–5 mm) followed by small fragments
III	Large fragments throughout

Clinical features

- Asymptomatic (silent obstruction), colic, nausea; fever and sepsis if infected; anuria and AKI (acute kidney injury) in a solitary kidney. Most lie in the distal ureter.

Evaluation

- KUB (kidney, ureter, bladder) X-ray, ultrasound, or non-contrast CT (computed tomography); creatinine, urine culture, blood counts.

Management

```
Steinstrasse
 ├─ infection / sepsis  → urgent PCN (percutaneous nephrostomy) + antibiotics
 ├─ asymptomatic, no obstruction → MET + analgesia, imaging every 1–2 weeks
 └─ failure, pain, obstruction:
      ├─ lead fragment (type II/III) → SWL to lead fragment
      └─ URS with laser lithotripsy (most definitive)
```

- **MET (medical expulsive therapy):** α-blockers (tamsulosin) improve fragment clearance and reduce pain.
- **PCN** is preferred in infected obstruction: drains without manipulating the ureter, and fragments often pass once the kidney is decompressed. A stent is an alternative.
- Retained fragments with poor progress: URS clears the column.

Prevention

- Choose PCNL or RIRS (retrograde intrarenal surgery) for large burdens.
- Energy ramping, slower shock rate; MET after SWL.
- **Stent before SWL:** may reduce steinstrasse with large stones but does not improve stone-free rate and adds stent symptoms. The chapters disagree: Chapter 97 says it may reduce steinstrasse; Chapter 98 advises against stenting at SWL.

Diagram to draw: ureter with lead fragment and trailing column (Coptcoat types I–III).

Key points to remember:

- Risk rises several-fold with burden >2 cm^2.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

Q3f. Renal aangiomyolipoma

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 3, Q3d)

Assumed meaning: renal angiomyolipoma (AML).

Introduction

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin) or TSC1 (hamartin) → unchecked **mTOR (mammalian target of rapamycin)** → growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis); treat as RCC (renal cell carcinoma).

Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock). CKD (chronic kidney disease) in TSC.

Diagnosis

- **CT (computed tomography): macroscopic fat (below about −15 to −20 HU [Hounsfield units]) on non-contrast images is diagnostic.**
- US (ultrasound) hyperechoic (so are some RCCs); MRI (magnetic resonance imaging): signal loss on fat suppression, India-ink artefact.
- Fat-poor lesion → MRI ± biopsy (HMB-45 [human melanoma black-45], Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

Management

Situation	Management
Acute haemorrhage	Resuscitate → **SAE (selective arterial embolisation)** → nephrectomy if SAE fails
Symptomatic	SAE, NSS (nephron-sparing surgery), ablation in selected cases
Asymptomatic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy**	Active treatment
None of these	**Surveillance**: imaging at 6 and 12 months, then yearly
TSC, asymptomatic, **>3 cm**	**Everolimus**; MRI every 1–3 years

- The old **4 cm** cut-off overtreats; it should not be a general trigger.

Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis), abscess, infarction.
- **Partial nephrectomy:** preferred surgery, essential in TSC (multifocal, recurrent).
- Thermal ablation: limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%**; stomatitis, acne, hyperlipidaemia.

Diagram to draw: management algorithm by bleeding, symptoms, size and TSC.

Key points to remember:

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Rapid Revision Sheet – Practice Set 7 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. How will you evaluate and manage 38 yr male with uncontrolled hypertension (LAQ)

- Hypertension below 40 years: screen for secondary causes.
- ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
- Metanephrines for phaeochromocytoma; α before β.
- FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Q2. Describe etiopathogenesis of stone formation and medical management of urolithiasis (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L; MET for distal 5–10 mm stones.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5, <250 mg/L.

Q3a. Discuss management of T1G3 bladder tumor (SAQ)

- Re-TURBT in all T1 at 2–6 weeks.
- BCG with 3-year maintenance reduces progression.
- Very high risk or BCG-unresponsive: radical cystectomy.

Q3b. Organ preservation solution (SAQ)

- Hypothermia slows, not stops, metabolism.
- UW: lactobionate, raffinose, starch; HTK: histidine, tryptophan, ketoglutarate.
- Machine perfusion reduces DGF.

Q3c. Intra op mgt of pheo. (SAQ)

- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.

Q3d. Saturation biopsy in CA prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Q3e. Steinstresse. (SAQ)

- Risk rises several-fold with burden >2 cm^2.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

Q3f. Renal aangiomyolipoma (SAQ)

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.